

Test report

Replaces the "Evaluation Report" item of the submission guidelines (this project has no GenAI component). All numbers were produced on **2026-10-09 from a fresh clone** of the repository (new folder, new virtualenv, new databases `healthcare_fresh` and `healthcare_fresh_test`) by following the README; nothing here is estimated.

1. How to run

```
pytest # backend, needs healthcare_test (README step 4); about 11 minutes
pytest --cov=src --cov-report=term-missing
cd frontend && npm test # 42 unit and component tests
node frontend/e2e/run.mjs # 71 real-browser checks (Chrome, API on :8000, dev server on :5173; E2E*_P
ASSWORD variables)
python scripts/verify_kpis.py # 88 checks: SQL = API = pandas on the raw CSV
```

Safety: the database fixture (`tests/conftest.py`) refuses to run unless `TEST_DATABASE_URL` names a schema ending in `_test` that differs from the development database, so a test can never clean up loaded data. One test reads `data/incoming/faulty_batch.json`, which exists after `python scripts/make_faulty_batch.py` (README step 11); without it that single test is skipped.

2. Results

Suite	Result
Backend (pytest) on the fresh clone	314 passed, 1 skipped in 11 min 36 s (the skip needs <code>faulty_batch.json</code> ; run on its own after <code>make_faulty_batch.py</code> , the file passes: 21 of 21)
Backend on the development checkout (earlier run, before the log-hygiene test was added)	314 passed in 10 min 12 s
Audit and log-hygiene change, targeted	99 passed (patients, audit, listing, log hygiene)
Frontend (Vitest) on the fresh clone	42 passed, 8 files
Real-browser run (headless Chrome, three roles) on the fresh clone	71 of 71 checks passed
KPI cross-check (<code>verify_kpis.py</code>) on the fresh clone, all five batches	88 of 88 checks passed

Independence from leftover data. The suite passed from a completely empty database (the fresh clone) and repeatedly on the development test schema without manual cleanup; the loaded development database (101,766 encounters) was unchanged after the runs.

3. Coverage (line coverage of `src/`, `pytest-cov`)

94% overall: 2,475 statements, 146 missed. 40 of 57 files are at 100%. Files below 100%:

File	Cover	What is not covered by pytest
etl/profile.py	0%	the one-off profiling script that produced docs/data_profile.md (run by hand, output reviewed)
etl/pipeline.py	88%	the command-line entry point (main , configure_logging wiring); it is exercised by every real load, the hard-kill test and the Airflow runs, but not by pytest
app/services/encounters.py	98%	4 defensive branches
app/services/patients.py	95%	the 409-after-rollback branch (lines 56-58)
etl/extract.py	94%	three error branches of the ID-mapping parser
app/services/admin.py	94%	duplicate-username race branch
app/database.py , app/middleware.py , utils/logging_config.py , app/security.py , app/schemas/* , app/services/query_helpers.py , app/services/audit.py , etl/load.py , etl/quality.py , etl/reports.py	90-99%	single defensive lines

Coverage is not the goal. The rules that matter were checked **by name** (section 5): the eligible denominator, idempotency, rollback, soft delete, RBAC and audit atomicity each have tests that fail when the rule is broken.

4. Deliberate-bug checks (tests must fail when the code is wrong)

Mutation	Result
Every encounter counted as readmission-eligible (ELIGIBLE in analytics.py replaced by an always-true condition)	6 tests fail: summary hand calculation, summary equals headline SQL, readmissions by age group, by other groups, medications and utilisation
Race and gender copied back into the patient audit snapshot	2 tests fail: the log-hygiene test and the patient audit test
(Earlier) an unprotected route added	the "every route needs a token" test flags it
(Earlier) audit write or commit forced to fail	no data row and no audit row remain, for create, update and delete

Both files were restored byte-for-byte after the mutations (hash compared).

5. The 20 silent-failure traps of the guide (Appendix B)

#	Trap	Evidence (test or recorded manual check)
1	Pandas turns "None" into missing	test_read_batch_missing_markers_and_none_value , test_lab_none_survives_build_tables ; profile shows 96,420 / 84,748 would be lost
2	Rate over all encounters	test_summary_matches_hand_calculation , test_summary_equals_the_l17_headline_sql , test_expired_group_has_no_eligible_encounters_and_null_rate , verify_kpis.py (prints the wrong-denominator value), mutation in section 4

#	Trap	Evidence (test or recorded manual check)
3	Encounters counted as patients	hand-built fixture has one patient with three encounters; headline <code>unique_patients</code> checked in <code>test_headline_matches_pandas_on_fixture</code> and <code>verify_kpis.py</code>
4	Soft-deleted rows leak	<code>test_soft_delete_drops_headline_by_exactly_one_and_undo_restores</code> , <code>test_deleted_patient_hides_all_their_encounters</code> , <code>test_soft_deleted_rows_and_deleted_patients_never_count</code> , <code>test_soft_deleted_encounters_are_not_exported</code>
5	Simulated dates shown as real	<code>test_every_response_says_dates_are_simulated_and_names_the_denominator</code> ; UI banner in the Vitest dashboard test and the browser run
6	Reload or renamed copy double counts	<code>test_same_file_twice_is_skipped_duplicate</code> , <code>test_renamed_reordered_copy_skips_all_rows</code> ; fresh-clone repeat scan loaded nothing
7	INSERT IGNORE hides errors	not used anywhere in <code>src/</code> (searched); <code>test_first_run_loads_and_reconciles</code> , <code>test_reconciliation_failure_rolls_back</code>
8	Commit per chunk	<code>test_forced_failure_mid_load_leaves_nothing</code> ; manual hard-kill run recorded in <code>docs/decision_log.md</code> (L6)
9	Age bracket sorted as text	<code>test_derive_age_order</code> , <code>test_readmissions_by_age_group_in_clinical_order_with_small_n</code>
10	ICD-9 codes converted to numbers	<code>test_transform.py</code> asserts V57 and E909 survive; <code>test_dq06_icd_format_corrects_without_rejecting</code> ; ICD prefix search tests
11	"No" medication rows stored	<code>test_medications_dimension_and_no_rows_for_unprescribed</code>
12	Join fan-out inflates counts	<code>test_group_counts_always_add_up_to_the_filtered_total</code>
13	Unstable paging, unbounded page size, free-text sort	<code>test_paging_walks_every_row_once_in_the_requested_order</code> ; parametrised invalid sort and page-size cases in <code>test_listing.py</code> (including <code>encounter_id;DROP TABLE x</code>)
14	Function on an indexed column	<code>test_indexes.py</code> (indexes exist, apply is idempotent); EXPLAIN plans naming each key in <code>docs/performance_notes.md</code> (manual evidence)
15	Stale responses overwrite newer filters	Vitest "ignores a slow response for an OLD filter that arrives after the newest one"; browser run changes the filter six times in half a second and compares with the API
16	Download with <code>window.open</code>	browser check "Excel report downloads through the UI with authentication (blob, no token in URL)"
17	Hiding buttons treated as security	<code>docs/role_verification.md</code> (UI versus direct API call per role and action, all agree); <code>test_wrong_role_is_403_right_role_is_200</code> , <code>test_role_matrix_for_patients</code> , <code>test_permission_matrix_in_code_matches_the_contract_document</code>
18	Airflow installed in the project venv	separate virtualenv in WSL2 (<code>docs/runbook.md</code>); <code>test_dag_files_compile_and_are_thin</code> ; the API suite passed after the install
19	Secrets or raw data committed	<code>.gitignore</code> written first; before the first commit the real database password, JWT secret and MySQL root password were searched in every staged file (0 hits); no CSV other than the lookup, the 500-row sample and a 30-row fixture is tracked (manual check, 2026-10-09)

#	Trap	Evidence (test or recorded manual check)
20	Tests pointed at the development database	fixture refuses non- *_test schemas (tests/conftest.py); the development row count was identical before and after the suite (manual check)

6. Logs, secrets and error responses

- **Scan of all log files** (logs/ , about 1.7 MB) for JWTs, bearer tokens, password= values, bcrypt hashes, JWT_SECRET: **0 matches**. race or gender appear only in profile.log , the L1 profiling script's aggregate value counts of the raw file (no per-patient rows).
- tests/api/test_log_hygiene.py drives good and bad logins, patient create/update/delete, invalid bodies, a forbidden call, an export and a forced 500 while capturing every log record at DEBUG; it fails if any password, token, hash, race value or demographic field name appears in the logs, the whole audit_logs table or any error body.
- **Finding fixed during this lab:** patient audit rows stored the before and after values of race and gender. The audit now records only that demographics are set and which field changed (changed_fields), never the values; the old synthetic audit rows in the development database were scrubbed. A test pins the new behaviour and the mutation in section 4 proves it.
- **Stack traces and SQL:** every error uses the single contract body; a forced ZeroDivisionError returns a generic 500 with no exception name, traceback or file/line; a database outage returns a generic message (test_errors.py). No endpoint was found that can return SQL text: SQL is built with bound parameters and sort columns come from a whitelist.
- **Swallowed exceptions and prints:** none of the latter in src/ ; the single broad except (in pipeline.py) is deliberate: it logs the full traceback with log.exception , closes the run as FAILED with the error message, loads nothing, and the command-line exit code becomes 1, so the failure is neither hidden nor left as a RUNNING row.

7. Missing tests ranked by risk (and what was done)

Risk	Gap	Status
HIGH	Secrets or demographics in logs, audit trail or errors	Added (test_log_hygiene.py); found and fixed the audit leak
HIGH	Eligible-denominator regression	covered by six tests; mutation proves it
MEDIUM	etl.pipeline.main (CLI path) has no pytest test	open; exercised by the hard-kill test, every real load and Airflow runs
MEDIUM	Guard that the test fixture refuses a non-test database	open; manual check recorded above
LOW	etl/profile.py untested	open; one-off script with reviewed output
LOW	Concurrent duplicate create (two requests at once)	open; the 409 branch is covered sequentially
LOW	Accessibility, tablet layout, screen reader	open; not tested
LOW	Opening exported .xlsx in Excel itself	open; files are validated with openpyxl only
LOW	Airflow 2.x compatibility	open; only 3.1 was run (the DAGs fall back to 2.x imports but this was not tested)

Risk	Gap	Status
LOW	Load test beyond 100k rows	open; performance measured with 101,766 encounters plus 300,000 audit rows only

8. What is mocked and what is not

Nothing in the database layer is mocked: loaders, analytics, RBAC, audit and soft delete are tested against the real MySQL test schema, because a mocked database passes even when the SQL is wrong. The Vitest component tests mock only the API client; the browser run then checks the real UI against the real API.

9. Known gaps

Section 7 lists them. In short: no real tablet or screen-reader test; the CLI entry point and the profiling script are outside pytest; Excel itself was not used to open the reports; Airflow was verified on 3.1 in WSL2 only and triggered from the command line rather than by hand in the UI.